

- 6 Students are learning about a simple check digit method for data validation. In this method, a single check digit is appended to the end of an original number to give a new number.

The students are studying a method which:

- calculates the sum of all the digits in the original number
- uses integer division to calculate the remainder when the sum is divided by 10
- uses the remainder as the check digit
- appends the check digit to the original number, creating the new number.

For example:

original number	4162
sum of all digits	$4 + 1 + 6 + 2 = 13$
remainder when the sum is divided by 10 using integer division	3
new number	41623

The original number is always at least three digits in length.

When the new number is input, the check digit is used to validate the new number.

- (a) A function `Generate()` is written to take an original number as a parameter and to return the check digit.

Outline a test plan that could be used to fully test function `Generate()`

Assume that the parameter is valid.

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